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Two magmatic events within the southern Indochina block constrained by new zircon U-Pb age data, SW-Vietnam and SE-Cambodia

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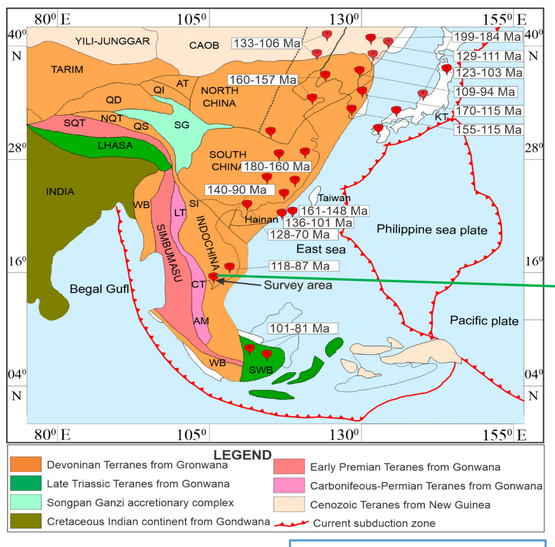
MinPet 2019

Introduction

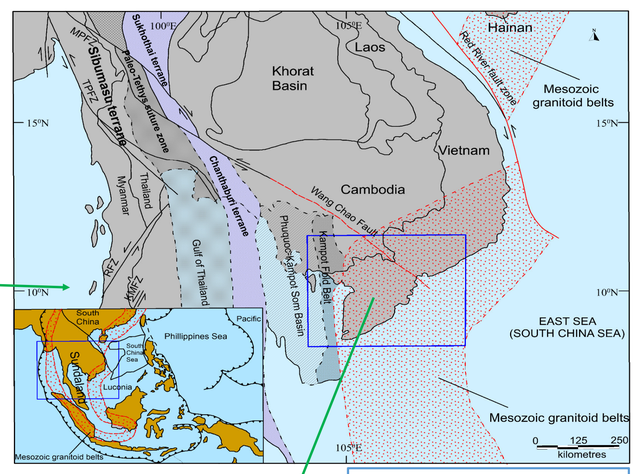
Magmatism occurs in SW Vietnam and SE Cambodia (SEVN-SEKH) from the Late Paleozoic to the Late Mesozoic which is related to two significant tectonomagmatic periods: (1) the opening and closure of the Tethyan Ocean and (2) the subduction of the Paleo-Pacific Ocean beneath the Eurasian continent. Two main distinct groups of Mesozoic igneous rocks corresponding to both tectonic events were found in the SWVN-SEKH based on zircon U-Pb ages.

- ✓The igneous rock suite related to the first event (Honkhoai-Tamao suite) displays an Early Mesozoic age (ca. **210-180 Ma**) and is mainly composed of granodiorite dispersedly outcropping at small islands in southernmost Vietnam.
- ✓The latter event resulted in the emplacement of voluminous basic to felsic rocks (i.e. Dinhquan, Deoca-Phnom Den, Ankroet-Ba Phnom, and Pha Aok suites) in the SWVN-SEKH area during the Late Mesozoic (ca. **108-78 Ma**) which occupy much of the studied area.

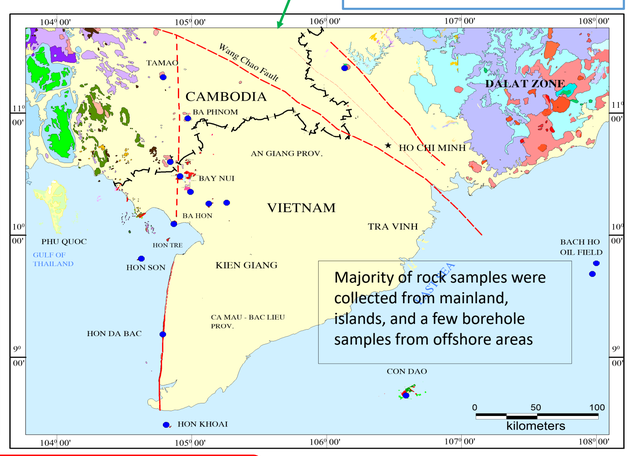
Geological setting and fieldwork



Modified after Metcalfe (2017)

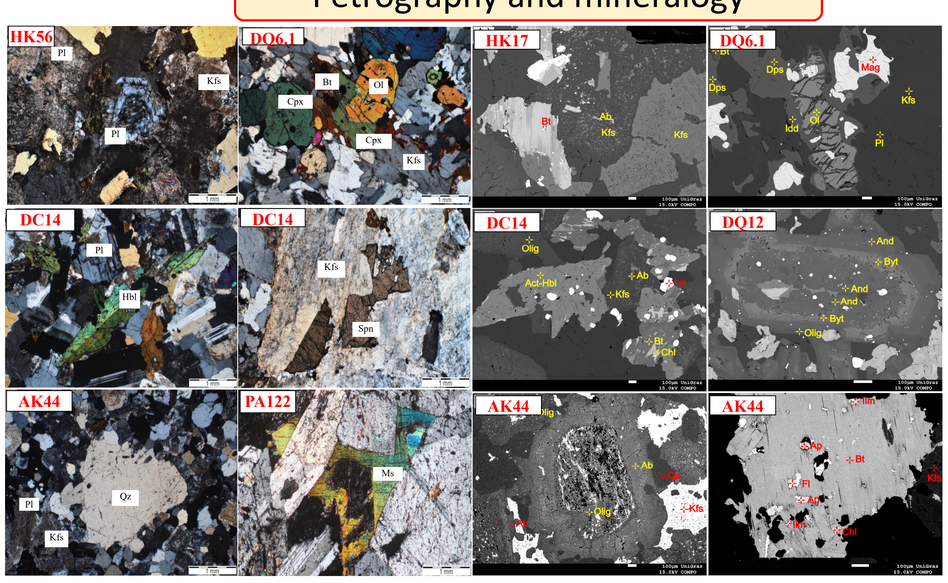


Modified after Fyhn et al. (2016)



- The studied area is around 100 000 km^2 in size, located in the southern Indochina block. It borders the Dalat zone to the north, which contains a large number of Late Mesozoic granitoid intrusions.
- A Jurassic-Cretaceous arc-related igneous belt runs from **Japan to southern Borneo** (Zhou and Li, 2000; Thuy et al., 2004).
- The Wang Chao fault is situated parallel to the Red River Fault and cause obvious separation in the young geological structures of the Indochina block.

Petrography and mineralogy

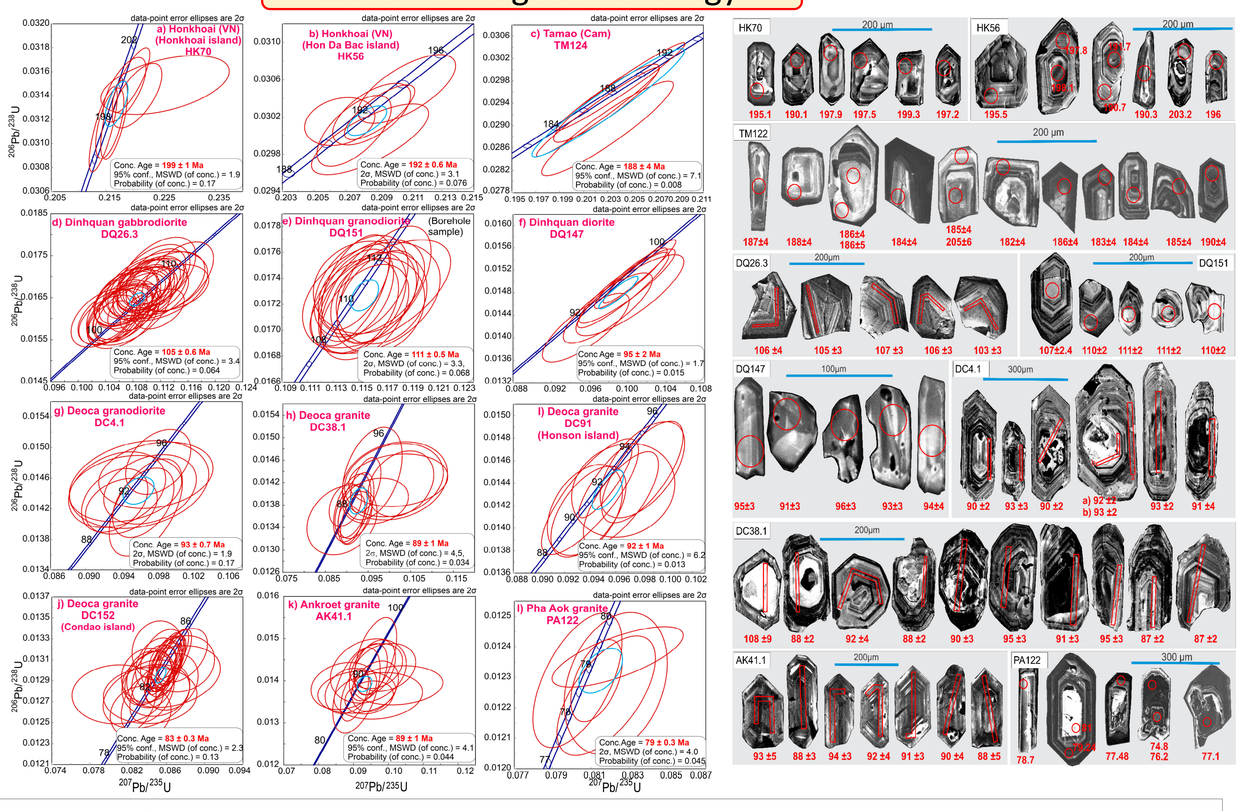


Cross-polarized light and backscattered electron images of minerals, olivine (Ol), clinopyroxene (Cpx), plagioclase (Pl), hornblende (Hbl), K-feldspar (Kfs), biotite (Bt), muscovite (Ms) quartz (Qz), sphene (Spn), diopside (Dps), plagioclase (Pl), andesine (And), bytownite (Byt), oligioclase (Olig), albite (Ab), magnetite (Mag), iddingsite (Idd), apatite (Ap), sphene (Spn), actinolite (Act), ilmenite (Ilm), fluorite (Fl), chlorite (Chl)

Five groups of igneous rocks were found in the area:

- Honkhoai-Tamao suite (granodiorite and granite)
- Dinhquan suite (gabbro-diorite, syenite, monzogabbro-diorite, and granodiorite)
- Deoca-Ba Phnom suite (pinkish granodiorite & granite)
- Ankroet-Baphnom suite (leucocratic fine-grained granite)
- Pha Aok suite (two mica-bearing coarse-grained granite)

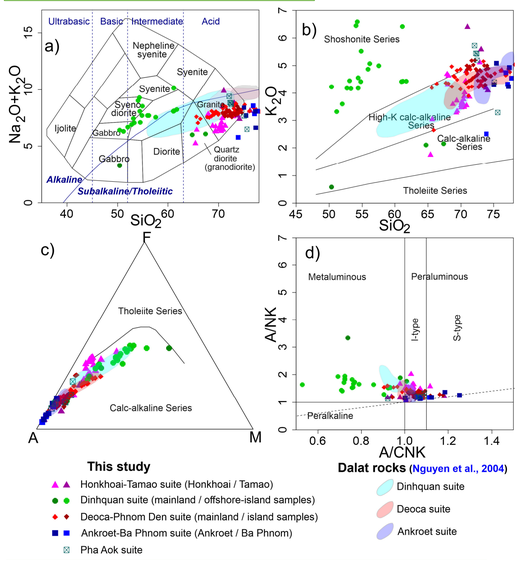
Zircon U-Pb geochronology



- Honkhoai-Tamao suite reveals a S-N trend in ages where the rocks become younger towards the north.
- Dinhquan, Deoca-Phnom Den, Ankroet-Ba Phnom, and Pha Aok rocks display a SE-NW trend where their ages become gradually younger from the "convergent margin" (offshore) to the "within-plate" (mainland) area.

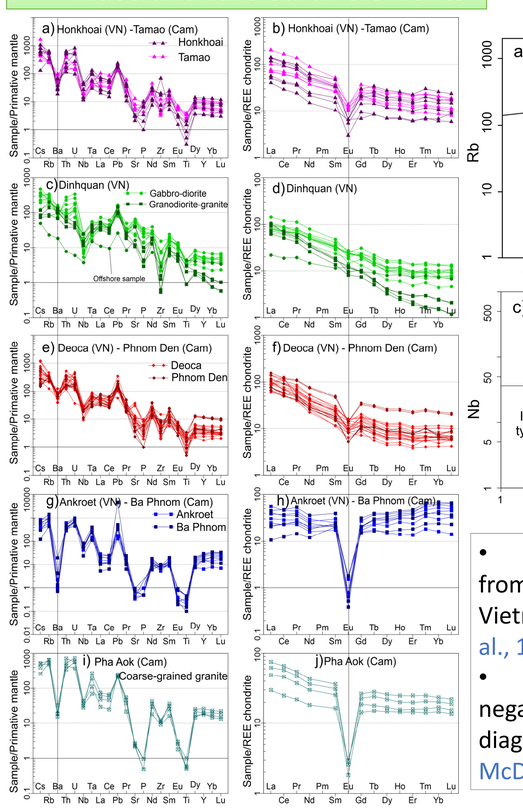
Geochemistry

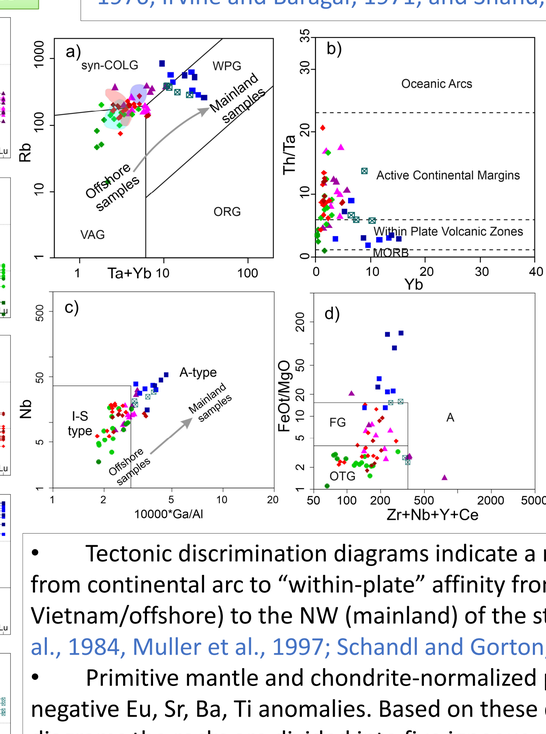
- Major elements



- The Dinhquan suite is categorized as alkaline and metaluminous rocks, while the other rock suites are subalkaline and from metaluminous to peraluminous.
- In comparison with igneous rocks from the Dalat zone, the studied rocks are wider in composition range and display higher content of potassium (Cox et al., 1979), Peccerillo and Taylor, 1976; Irvine and Baragar, 1971; and Shand, 1943)

- Trace and rare earth elements





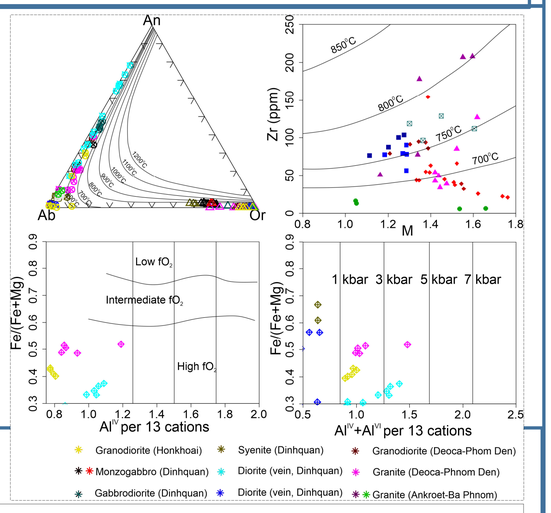
The Honkhoai-Tamao, Dinhquan, and Deoca-Phnom Den suite rocks exhibit I-type affinity while the Ankroet-Ba Phnom and Pha Aok rocks are respectively compatible with A2-type and S-type granitoids (Whalen et al . 1987 and Eby, 1992).

- The rocks present a wide range from ordinary to fractionated igneous rocks.

- Tectonic discrimination diagrams indicate a magmatic evolution from continental arc to "within-plate" affinity from the SE (coastal SE-Vietnam/offshore) to the NW (mainland) of the study area (Pearce et al., 1984, Muller et al., 1997; Schandl and Gorton, 2002).
- Primitive mantle and chondrite-normalized patterns exhibit negative Eu, Sr, Ba, Ti anomalies. Based on these discrimination diagrams the rocks are divided into five igneous groups (Sun and McDonough, 1995; Nakamura, 1974)

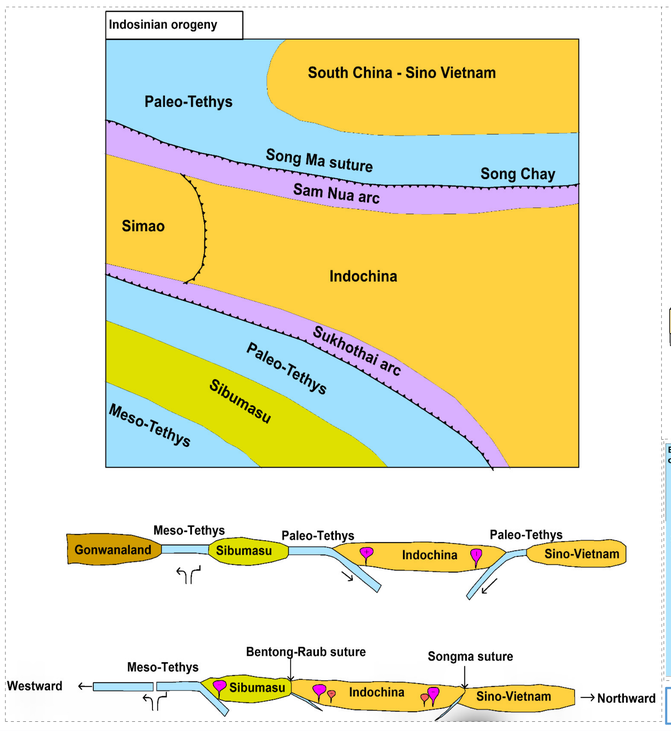
Thermobarometry

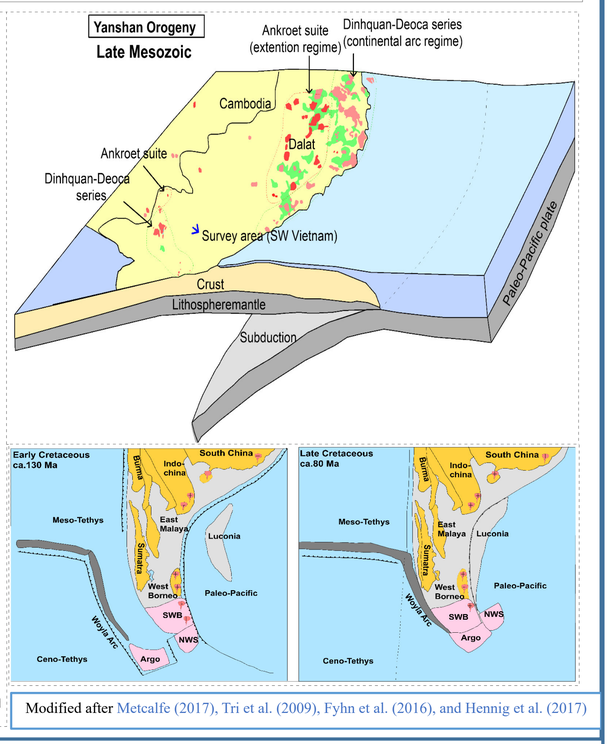
Thermobarometry was evaluated based on the chemical composition of rock-forming minerals (i.e., feldspar and Ca-amphibole) and whole-rock chemistry. The crystallization temperature of the rocks varies between min. 650 and 920 ^{0}C and the pressure range from below 1 kbar to 4 kbar. All of the studied rocks display a high fO_2 (Elkins and Grove, 1990; Watson and Harrison, 1983; Leake, 1978; Schmidt, 1992)



Geodynamic model and conclusion

- The Honkhoai-Tamao suite was emplaced coincidently with the Early Mesozoic magmatism which is linked to the Indosinian Orogeny driven by accretion of the Sibumasu terrane to the Indochina–South China block.
- The Dinhquan, Deoca-Phnom Den, Ankroet-Ba Phnom, and Pha Aok suites are related to the subduction event which existed along the eastern margin of the Eurasian continent during the Late Mesozoic, referred to as the Late Yanshan Orogeny.





Modified after Metcalfe (2017), Tri et al. (2009), Fyhn et al. (2016), and Hennig et al. (2017)

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